

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Programming Fundamentals and Data Structures	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated: 5th March, 2026
Semester: 1st	Session: 2026
Lab/Project/Assignment #: 4	CLOs to be covered: CLO1
Lab Title: Nested Loops	Teacher Name: Hafiz Muhammad Mubashar

Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Mark s	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 4: Nested Loops

Lab Goals/Objectives:

To understand and apply nested loops (a loop inside another loop) to solve pattern-printing and multi-dimensional iteration problems.

Equipment Required:

Python 3, VS Code / PyCharm

Theory:

A nested loop is a loop that runs inside the body of another loop. The inner loop completes all of its iterations for each single iteration of the outer loop, which makes nested loops useful for generating patterns and multiplication tables.

Lab Work:**Task 1: Nested Loop**

```
for i in range(1, 3):  
    for j in range(1, 3):  
        print("i=", i, "j=", j)
```

Result:

```
i= 1 j= 1  
i= 1 j= 2  
i= 2 j= 1  
i= 2 j= 2
```

Task 2: Coordinate Pattern

```
for i in range(4):  
    for j in range(4):  
        print(f"{i},{j}")
```

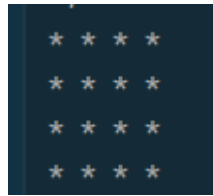
Result:

```
0,0  
0,1  
0,2  
0,3  
1,0  
1,1  
1,2  
1,3  
2,0  
2,1  
2,2  
2,3  
3,0  
3,1  
3,2  
3,3
```

Task 3: Square Pattern

```
for i in range(4):  
    for j in range(4):  
        print("*",end=" ")  
    print()
```

Result:



Task 6: Square Pattern (5x5)

```
for i in range(1,6):  
    for j in range(1,6):  
        print("*",end=" ")  
    print()
```

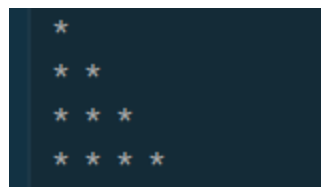
Result:



Task 4: Left-Aligned Triangle

```
for i in range(1,5):  
    for j in range(i):  
        print("*",end=" ")  
    print()
```

Result:



Task 5: Number Triangle

```
for i in range(1,5):  
    for j in range(i):  
        print(j,end=" ")  
    print()
```

Result:

```
0  
0 1  
0 1 2  
0 1 2 3
```

Task 6: Number Triangle (Inclusive)

```
for i in range(1,5):  
    for j in range(i+1):  
        print(j,end=" ")  
    print()
```

Result:

```
0 1  
0 1 2  
0 1 2 3  
0 1 2 3 4
```

Task 7: Number Triangle (Starting at 1)

```
for i in range(1,5):  
    for j in range(1,i+1):  
        print(j,end=" ")  
    print()
```

Result:

```
1  
1 2  
1 2 3  
1 2 3 4
```

Task 8: Number Triangle (Starting at 0)

```
for i in range(1,5):  
    for j in range(0,i+1):
```

```
    print(j,end=" ")
print()
```

Result:

```
0 1
0 1 2
0 1 2 3
0 1 2 3 4
```

Task 9: Number Triangle (Starting at -1)

```
for i in range(1,5):
    for j in range(-1,i+1):
        print(j,end=" ")
    print()
```

Result:

```
-1 0 1
-1 0 1 2
-1 0 1 2 3
-1 0 1 2 3 4
```

Task 10: Multiplication Table (1-10)

```
for i in range(1,11):
    for j in range(1,5):
        print(i*j,end=" \t")
    print()
```

Result:

```
1 2 3 4
2 4 6 8
3 6 9 12
4 8 12 16
5 10 15 20
6 12 18 24
7 14 21 28
8 16 24 32
9 18 27 36
10 20 30 40
```

Task 11: Multiplication Table (Transposed)

```
for i in range(1,5):
    for j in range(1,11):
        print(i*j,end=" \t")
    print()
```

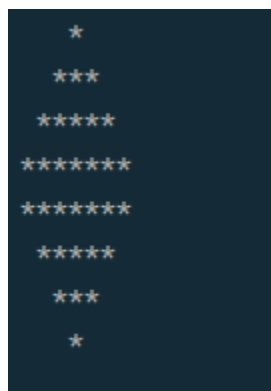
Result:

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	9	12	15	18	21	24	27	30
4	8	12	16	20	24	28	32	36	40

Task 12: Diamond Pattern

```
for i in range(1,5):
    for j in range(4-i):
        print(" ",end="")
    for j in range(2*i-1):
        print("*",end="")
    print ()
for i in range (4,0,-1):
    for j in range(4-i):
        print (" ",end="")
    for j in range(2*i-1):
        print("*",end="")
    print ()
```

Result:



Conclusions:

Nested loops were used extensively to generate number/star patterns and multiplication tables, reinforcing control over multi-level iteration.